

Topic 5 Review [99 marks]

1. [Maximum mark: 15]

25M.1.HL.TZ2.12

(a) Consider the following recursive function:

```
recfn(N)
  if N <= 0
    then
      return 0
    else
      if N mod 2 = 0
        then
          return recfn(N-1) - N
        else
          return recfn(N-1) + N
      endif
    endif
  end recfn
```

Determine the value of variable X after execution of the following program statement:

$X = \text{recfn}(5)$

Show all your working.

[5]

(b) Describe **two** basic stack operations.

[4]

The **NUMBERS** stack stores real numbers. An algorithm should be constructed that will find the smallest number and the largest number stored in the **NUMBERS** stack. After execution of the algorithm, **only** these two numbers should be stored in the **NUMBERS** stack (see **Figure 1**).

Figure 1: Example data stored in the NUMBERS stack before and after execution of the requested algorithm

The content in the **NUMBERS** stack **before** execution of the algorithm.

The number **23.0** is on the top of the **NUMBERS** stack.

23.0
-54.9
-12.4
67.0
15.2

The content in the **NUMBERS** stack **after** execution of the algorithm.

The smallest number, **-54.9**, is on the top of the **NUMBERS** stack.

-54.9
67.0

- (c) Construct this algorithm in pseudocode.

You may assume that the **NUMBERS** stack already contains real numbers.

You must use the access methods of a stack.

[6]

2. [Maximum mark: 4]

24N.1.HL.TZ0.7

Consider the following recursive method:

```
rec(A,B)
  if B=0
    then
      return A
    else
      return rec(B,A mod B)
  end if
end rec
```

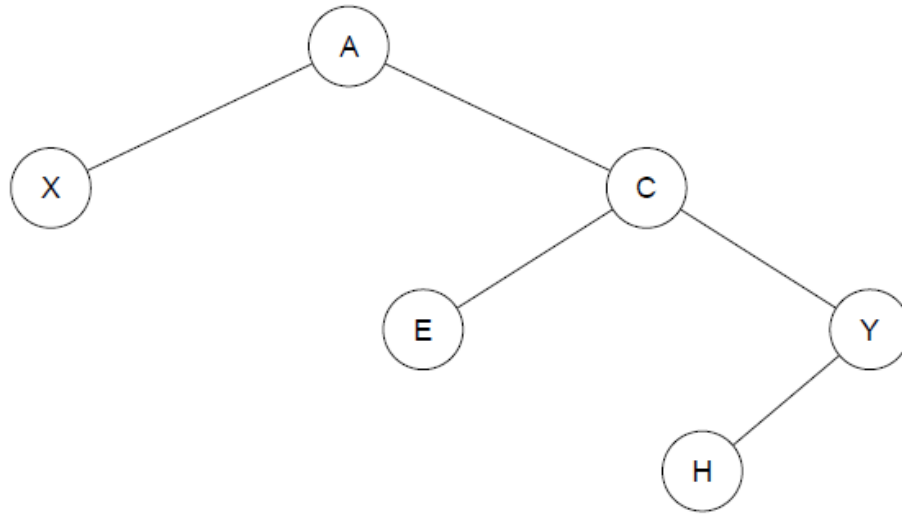
Determine the value of **rec(20,12)**. Show all your working.

[4]

3. [Maximum mark: 2]

24N.1.HL.TZ0.9

Consider the following binary tree:



- (a) State the result of postorder traversal of the binary tree. [1]
- (b) State the result of inorder traversal of the binary tree. [1]

4. [Maximum mark: 4] 24N.1.HL.TZ0.10
Outline **two** operations on a queue data structure. [4]

5. [Maximum mark: 15] 24N.1.HL.TZ0.13
Cell phones are used to make phone calls and send text messages. They can also browse the internet and run apps.

An incoming call will interrupt an app being executed.

- (a) Describe how a stack data structure can be used to resume the app that was initially running. [2]
- (b) Describe **one** way a cell phone app can help the user improve their health. [2]

- (c) Outline why a higher screen resolution leads to higher quality images on a cell phone screen. [2]
- (d.i) Explain what is meant by GPS (global positioning system). [4]
- (d.ii) Outline **one** concern of cell phone apps sharing the location of the user. [2]
- (e) Explain **one** advantage of using a dedicated operating system on a cell phone instead of a generic operating system. [3]

6. [Maximum mark: 15] 24N.1.HL.TZ0.14

An application requires a list of students' names to be held in alphabetical order, from A to Z, in the primary memory.

The application must allow for insertion and deletion of names.

- (a) Outline **two** advantages of using a linked list rather than a static array in this scenario. [4]
- (b) Explain how a specified student's name could be found in a linked list. [5]

A double linked list would be preferred over a single linked list for applications that require the ability to move forward (from A to Z) and backward (from Z to A) through the list of names.

- (c.i) Describe the structure of a double linked list. [4]
- (c.ii) State **two** applications of double linked lists in computing. [2]

7. [Maximum mark: 15] 24N.1.HL.TZ0.15

The two-dimensional array **MAT** is used to store randomly generated integers in the range from 0 to 9.

In the **MAT** array, some numbers can be repeated multiple times and other numbers may not appear at all (see **Figure 1**).

Figure 1: Example data stored in the MAT array

	[0]	[1]	[2]	[3]	[4]	[5]	[6]
[0]	0	2	5	9	3	6	1
[1]	0	2	9	0	7	0	0
[2]	4	2	7	9	5	7	1
[3]	7	2	3	1	5	4	0
[4]	3	6	3	9	9	6	0
[5]	7	4	3	1	7	9	3
[6]	3	9	6	4	2	2	4

The one-dimensional **COUNT** array will show how many times each of the randomly generated numbers is found in the **MAT** array (see **Figure 2**).

Figure 2: The content of the COUNT array

[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
7	4	6	7	5	3	4	6	0	7

Figure 2 shows that number 4 occurs 5 times (**COUNT**[4]=5) in the **MAT** array.

- (a) Construct an algorithm in pseudocode to create the **COUNT** array as described. You may assume that the **MAT** array is created and the **COUNT** array is initialized with zero values.

[4]

In **Figure 2**, **COUNT**[8]=0 because the number 8 does not appear in the **MAT** array.

- (b) Construct an algorithm in pseudocode that uses the **COUNT** array to output the numbers that do not appear in the **MAT** array. If all of the numbers from 0 to 9 are present, then it will output an appropriate message. [5]

The mode is the number that appears most often in the **MAT** array. There can be more than one mode.

0, 3 and 9 are the modes of the numbers stored in the **MAT** array (see **Figure 2**).

- (c) Construct an algorithm in pseudocode that uses the **COUNT** array to output the mode(s) of the numbers stored in the **MAT** array. [6]

8. [Maximum mark: 2] 24M.1.HL.TZ1.8
Identify **two** applications of a stack. [2]

9. [Maximum mark: 15] 24M.1.HL.TZ1.15
Linked lists and binary trees are examples of dynamic data structures.

- (a.i) Outline **one** benefit of using dynamic data structures. [2]
(a.ii) Outline **one** drawback of using dynamic data structures. [2]

Figure 3 represents a linked list containing the numbers 3, 11, 19 and 31.

Figure 3: A linked list



- (b) Describe the steps to find and delete the node containing data item 19.

[5]

(c) Describe the structure of a node in a binary tree.

[2]

These numbers are input in the following order:

17 11 23 5 3 19 37 31

and inserted in a binary tree such that an inorder traversal of the binary tree outputs the numbers sorted in ascending order.

(d.i) Sketch the resulting binary tree.

[3]

(d.ii) State all the leaf nodes in the binary tree sketched in part (d)(i).

[1]

10. [Maximum mark: 2]

23M.1.HL.TZ1.2

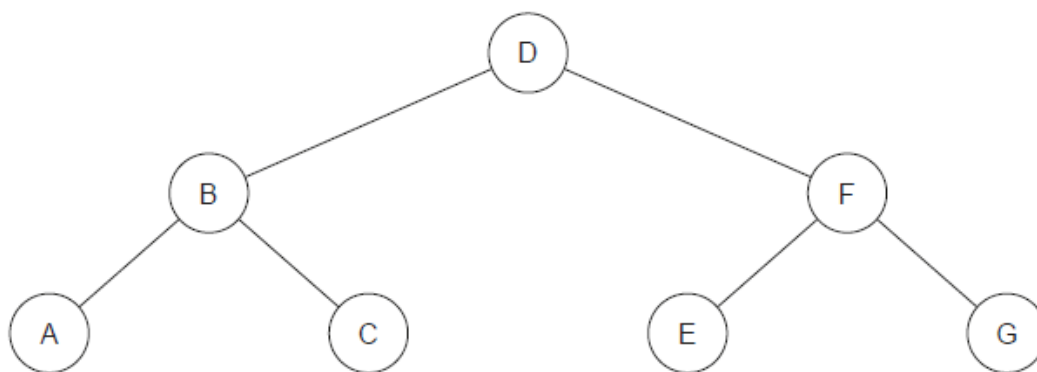
Identify **two** applications of queues in computing.

[2]

11. [Maximum mark: 5]

23M.1.HL.TZ1.8

Consider the following binary tree, in which each node stores a value greater than all the values in the node's left subtree and less than those in its right subtree.



(a) Identify the leaf nodes in this binary tree.

[1]

- (b) State the result of the postorder traversal. [1]
- (c) Sketch the resulting binary tree after the deletion of the root node. [3]

12. [Maximum mark: 3] 22N.1.HL.TZ0.8

A computer science student is coding and running a program while several documents, such as essays, lab reports and homework, are being printed out.

- (a) Define the term *queue* as a data structure. [1]
- (b) Identify **two** different queues that are used in this scenario. [2]

13. [Maximum mark: 2] 22M.1.HL.TZ0.9

- (a) Identify **one** characteristic of a queue. [1]
- (b) Identify **one** application of a queue. [1]